Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Unphysicality of Hilbert spaces for quantum mechanics

Files.

- File: `briefs/HilbertUnphysicality/HilbertUnphysicality.lean`
- Library: `HilbertUnphysicality`
- Imports: `InconsistentConvergence`, one new `ForPhyslib` file:
  `ForPhyslib/QuantumMechanics/UnitaryEquivalence.lean` (a unitary carries the quantum model),
  the existing `ForPhyslib/QuantumMechanics/Stone.lean` (Stone's theorem in Schrödinger form, extended here by the differentiable orbits),
  three TauCeti files, the Hermite functions, their Hilbert basis of $L^2(\mathbb{R})$ and their Schwartz versions:
  `TauCeti/Analysis/SpecialFunctions/Hermite/Function/HilbertBasis.lean`,
  `TauCeti/Analysis/SpecialFunctions/Hermite/Function/Oscillator.lean`,
  `TauCeti/Analysis/SpecialFunctions/Hermite/Function/Schwartz.lean`,
  and thirteen new `ForMathlib` files:
  `ForMathlib/Analysis/InnerProductSpace/l2Diagonal.lean` (diagonal operators on ℓ^2,

and their characterization by eigenvectors),
`ForMathlib/Analysis/InnerProductSpace/l2DiagonalEvolution.lean` (their evolution),
`ForMathlib/Analysis/SpecialFunctions/Gaussian/Erf.lean` (the error function),
`ForMathlib/Analysis/InnerProductSpace/l2Space.lean` (separable Hilbert spaces are $\ell^2(\mathbb{N})$),
`ForMathlib/MeasureTheory/Function/L2Space.lean` (L^2 is infinite-dimensional),
`ForMathlib/MeasureTheory/Function/L2Multiplication.lean` (multiplication operators on L^2, their self-adjointness, bounds and evolution),
`ForMathlib/MeasureTheory/Function/LpSpace/ChangeOfVariables.lean` (the unitary of a change of variables on $L^2(\mathbb{R})$, and how it transforms multiplication operators),
`ForMathlib/LinearAlgebra/LinearPMap.lean` (composition on the natural domain),
`ForMathlib/Analysis/InnerProductSpace/LinearPMap.lean`
(symmetric extensions of self-adjoint operators, bounded symmetric perturbations, and closed operators under summation),
`ForMathlib/Analysis/InnerProductSpace/Subspace.lean` (summable combinations of an orthonormal family),
`ForMathlib/Analysis/Normed/Algebra/Wielandt.lean` (the Wielandt–Wintner theorem),
`ForMathlib/Analysis/Distribution/TemperedDistribution.lean` (the Leibniz rule in $\mathcal{S}'$),
`ForMathlib/MeasureTheory/Measure/Typeclasses/NullSingletonClass.lean` (changing a function at a point)

- Second file, for potentials in $L^2 + L^\infty$:
`briefs/HilbertUnphysicality/HilbertUnphysicality/KatoRellich.lean`
(module `HilbertUnphysicality.KatoRellich`). It imports the main file and two new `ForMathlib` files:
`ForMathlib/Analysis/InnerProductSpace/KatoRellich.lean`
(TauCeti's Kato–Rellich theorem, behind a private import),
`ForMathlib/Analysis/Fourier/LpSpace.lean` (the Fourier transform on L^2 of integrable functions, and the L^1 bound on $\mathbb{R}$).
It is kept out of the comparator lane: see "Differences".

- Third file, for real potentials:
`briefs/HilbertUnphysicality/HilbertUnphysicality/RealPotential.lean`
(module `HilbertUnphysicality.RealPotential`). It imports the main file, the Fourier file above, which now also has the Fourier transform of a conjugate, and two new `ForMathlib` files:
`ForMathlib/Analysis/InnerProductSpace/SelfAdjointExtension.lean`
(von Neumann's theorem on self-adjoint extensions, with TauCeti's criterion behind a private import),
`ForMathlib/MeasureTheory/Function/L2Conj.lean` (complex conjugation on L^2).
It is kept out of the comparator lane for the same reason.

- Fourth file, for Schwartz spaces:
`briefs/HilbertUnphysicality/HilbertUnphysicality/SchwartzSpaces.lean`
(module `HilbertUnphysicality.SchwartzSpaces`). It imports the main file and five new `ForMathlib` files, in `ForMathlib/Analysis/Distribution/SchwartzSpace/`:
`Hermite.lean` (the Hermite functions h_α on $\mathbb{R}^n$ as Schwartz functions, their Hilbert basis of $L^2(\mathbb{R}^n)$ and the ladder relations for the coefficients),

`HermiteCoeff.lean` (the coefficients of a Schwartz function decay rapidly),
`HermiteSeminorm.lean` (the Schwartz seminorms are bounded by the coefficients),
`HermiteSeries.lean` (rapidly decreasing coefficients sum to a Schwartz function),
`HermiteEquiv.lean` (the isomorphism $\mathcal{S}(\mathbb{R}^n) \cong \mathcal{S}(\mathbb{R}^m)$),
with the existing `ForMathlib/Analysis/Distribution/SchwartzSpace/SupNorm.lean` (a Sobolev bound for the values) and four TauCeti files:
`TauCeti/Analysis/SpecialFunctions/Hermite/Function/Pi/Basis.lean`,
`TauCeti/Analysis/SpecialFunctions/Hermite/Function/Ladder.lean`,
`TauCeti/Analysis/SpecialFunctions/Gaussian/Schwartz.lean`,
`TauCeti/Analysis/InnerProductSpace/Parseval.lean`.
It is kept out of the comparator lane, which stays unchanged.

- Fifth file, for classical cutoffs:
  `briefs/HilbertUnphysicality/HilbertUnphysicality/ClassicalCutoffs.lean`
  (module `HilbertUnphysicality.ClassicalCutoffs`). It does not import the main file. It imports two new `ForMathlib` files:
  `ForMathlib/MeasureTheory/Measure/ProbabilityMeasure/CompactSupport.lean`
  (measures concentrated on a compact set: integrability, continuity of expectations, images),
  `ForMathlib/Analysis/Fourier/PaleyWiener.lean` (a function and its Fourier transform cannot both have bounded support),
  the second with the TauCeti file `TauCeti/Analysis/Fourier/SineSeries.lean` (uniqueness of Fourier coefficients on an interval).
  It is kept out of the comparator lane, which stays unchanged.

Setting. Theorem 5 says that three physical conditions are not guaranteed over the Hilbert-space formulation QSTO:H. The brief proves four mathematical propositions (Sec. 2) and then argues each part of Theorem 5 from one of them (Sec. 4). The propositions are checked in Lean, and Theorem 5 is stated and proved as `theorem5`, with each condition in its minimum form (see the open problems, "Precise readings"). Of the additional remarks of Sec. 5, the one on the domain of the Schrödinger equation is checked too, for every self-adjoint Hamiltonian and in detail for the harmonic oscillator, in the number basis and in position space, and for the free particle in position space, and so are the ones on commutator relations, for position and momentum on $L^2(\mathbb{R})$, and on point-wise values, in $L^2(\mathbb{R})$ (see "Differences"). The closing remark of the proof of Theorem 5, that Schwartz spaces present the same problem, is checked as well, and so are its remarks that classical mechanics allows a cutoff on all variables and quantum mechanics does not, in the readings of the open problems.

Results.

Brief	Lean
Proposition 1	
$\mathcal{D}(A) \neq \mathcal{H}$ (Hellinger–Toeplitz)	`bounded_of_isSymmetric`, `not_isSymmetric_of_unbounded`, `domain_ne_top_of_unbounded`
A is not continuous on $\mathcal{D}(A)$	`not_continuous_of_unbounded`
Proposition 2	
N with domain $\mathcal{D}(N)$ is self-adjoint	`numberOp`, `coe_domain_numberOp`, `isSelfAdjoint_numberOp` (from `lp.isSelfAdjoint_diagonal`)
N is unbounded, so not continuous on $\mathcal{D}(N)$	`exists_lt_norm_numberOp`, `not_continuous_numberOp`
$\mathcal{D}(N)$, $E_N(x) = \langle x\|Nx\rangle / \|x\|^2$	`numberDomain`, `expN`, `expN_eq_inner`
$x_j \to e_0$ in $\mathcal{D}(N) \setminus \{0\}$, $E_N(x_j) = 1$, $E_N(e_0) = 0$	`expN_not_continuousWithinAt`, `not_continuousOn_expN` (sequence from the inconsistent-convergence brief)
Proposition 3	
erf, $\|\operatorname{erf}\| < 1$, erf $\to \pm 1$	`Real.erf`, `Real.abs_erf_lt_one`, `Real.tendsto_erf_atTop`
$f(x) = \tan(\frac{\pi}{2} \operatorname{erf} x)$ is increasing	`frameChange`, `strictMono_frameChange`
f is a bijection of $\mathbb{R}$, $g = f^{-1}$, $g' > 0$	`surjective_frameChange`, `frameIso`, `hasDerivAt_frameIso_symm`
$U\psi = \phi$, at the level of densities	`cauchyDensity_frameChange_mul_deriv`: $\|\phi(f(x))\|^2 f'(x) = \|\psi(x)\|^2$
the same, integrated	`arctan_frameChange`: $\arctan(f(x))/\pi = \operatorname{erf}(x)/2$
ψ normalized, $\psi \in \mathcal{D}(X)$	`integral_gaussDensity`, `integrable_sq_mul_gaussDensity`
ϕ normalized, $\phi \notin \mathcal{D}(Y)$	`integral_cauchyDensity`, `not_integrable_sq_mul_cauchyDensity`
the unitary U, $(U\psi)(y) = \psi(g(y))\sqrt{g'(y)}$	`frameUnitary` (from `Lp.compDiffeo`), `coeFn_frameUnitary`
$\mathcal{D}(X) = \mathcal{D}(Y) = \{\psi : x\psi \in L^2\}$	`positionDomain`
$\psi \in \mathcal{D}(X)$, $U\psi = \phi$, $U\psi \notin \mathcal{D}(Y)$	`gaussState_mem_positionDomain`, `frameUnitary_gaussState`, `frameUnitary_gaussState_not_mem`
the minimum form of [B]WDQ fails	`not_forall_frameUnitary_mem_positionDomain`
X and Y, multiplication by the coordinate, are self-adjoint	`positionOp`, `isSelfAdjoint_positionOp` (from `L2.isSelfAdjoint_mulOp`)
UXU^{-1} is multiplication by $g(y)$, with domain $U\,\mathcal{D}(X)$	`conjLinearPMap_frameUnitary_positionOp` (from `Lp.compDiffeo_mulOp`)
UXU^{-1} is self-adjoint, and $UXU^{-1} \neq Y$	`isSelfAdjoint_conjLinearPMap_frameUnitary_positionOp`, `conjLinearPMap_frameUnitary_positionOp_ne`

Brief	Lean
Proposition 4	
separable infinite-dimensional Hilbert spaces are isomorphic	`nonempty_linearIsometryEquiv_of_separable`
$L^2(\mathbb{R}^n)$ is separable and infinite-dimensional	`L2.separableSpace_euclidean`, `L2.not_finiteDimensional_euclidean`
$L^2(\mathbb{R}^{n+1}) \cong L^2(\mathbb{R}^{m+1})$	`L2_euclidean_iso`
Theorem 5, [B]STDIST	
a unitary induces bijections between rays, density operators, self-adjoint operators	`mapRays`, `isDensityOperator_conjStarAlgEquiv_iff`, `bijective_conjLinearPMap`, `isSelfAdjoint_conjLinearPMap_iff`
and preserves the Born rule and the entropy	`born_map`, `IsDensityOperator.vonNeumannEntropy_conjStarAlgEquiv`
$L^2(\mathbb{R}^{n+1})$ and $L^2(\mathbb{R}^{m+1})$ carry the same QSTO:H model	`carriesQSTO`, `exists_carriesQSTO`
Theorem 5	
over QSTO:H, [B]COBS, [B]WDQ and [B]STDIST are not guaranteed	`theorem5`
Sec. 5, the domain of the Schrödinger equation	
$\mathcal{D}(H) \neq \mathcal{H}$ for the oscillator	`domain_numberOp_ne_top`, `harmonicState_not_mem_numberDomain` ($x_n = 1/n$)
the evolution e^{-itH}, defined on all of $\mathcal{H}$	`numberEvolution` (from `lp.diagonalEvolution`)
on $\mathcal{D}(H)$ the Schrödinger equation holds	`hasDerivAt_numberEvolution` (from `lp.hasDerivAt_diagonalEvolution`)
off $\mathcal{D}(H)$ it holds at no time	`not_differentiableAt_numberEvolution`, `differentiableAt_numberEvolution_iff`
this evolution is that of Stone's theorem	`numberEvolution_eq_schrodingerPropagator`
the same for every self-adjoint H unbounded on its domain	`schrodinger_only_on_domain` (from `schrodingerPropagator`, `differentiableAt_schrodingerPropagator_iff`)
$V(x)\psi(x)$ need not be square-integrable	`not_memLp_sq_mul_frameUnitary_gaussState`, `frameUnitary_gaussState_not_mem`

Brief	Lean
Theorem 5, proof, last paragraph: Schwartz spaces	
the Hermite functions $h_\alpha(x) = \prod_i \psi_{\alpha_i}(x_i)$ are Schwartz functions, and a Hilbert basis of $L^2(\mathbb{R}^n)$	`SchwartzMap.hermitePi`, `SchwartzMap.hermitePiBasis`, `SchwartzMap.toLp_hermitePi`
ladder relations for the coefficients $c_\alpha(f) = \langle h_\alpha \vert f \rangle$	`SchwartzMap.hermiteCoeff_coordMulCLM`, `SchwartzMap.hermiteCoeff_coordDerivCLM`
the coefficients of $f \in \mathcal{S}$ decay rapidly, with bounds by seminorms	`SchwartzMap.exists_hermiteNorm_le_seminorm`
the seminorms of f are bounded by the coefficients	`SchwartzMap.exists_seminorm_le_hermiteNorm`
rapidly decreasing coefficients sum to a Schwartz function	`SchwartzMap.hermiteSeries`, `SchwartzMap.hermiteCoeff_hermiteSeries`, `SchwartzMap.exists_seminorm_hermiteSeries_le`
a unitary $L^2(\mathbb{R}^{n+1}) \to L^2(\mathbb{R}^{m+1})$ that restricts to $\mathcal{S}(\mathbb{R}^{n+1}) \cong \mathcal{S}(\mathbb{R}^{m+1})$	`SchwartzMap.exists_unitary_schwartzEquiv`
it carries the QSTO:H model, so "Schwartz spaces present the same problem"	`schwartz_stdist`, `nonempty_schwartzEquiv`

Brief	Lean
Theorem 5, proof: classical cutoffs	
a continuous function is integrable for a finite measure concentrated on a compact set	`MeasureTheory.integrable_of_isCompact`
on the probability measures concentrated on a compact set, expectations are continuous for weak convergence	`MeasureTheory.ProbabilityMeasure.continuousOn_integral_of_isCompact`
a continuous map keeps a measure concentrated on a compact set	`MeasureTheory.Measure.map_compl_image_eq_zero`
[B]COBS: with a cutoff on all variables, expectations are finite and continuous	`classical_cutoff_cobs`
[B]WDQ: changes of frame keep the cutoff, "which bound all moments"	`classical_cutoff_wdq`
if $f \in L^2$ vanishes outside a bounded interval and $\mathcal{F}f$ vanishes on a nonempty open set, then $f = 0$	`Real.ae_eq_zero_of_fourier_eqOn_zero`, `Real.ae_eq_zero_of_fourier_eq_zero_of_lt_abs`
"Quantum mechanics does not allow that": no nonzero state has a cutoff on position and momentum	`quantum_no_cutoff`

Brief	Lean
Sec. 5, in position space	
$P = -i\,d/dx$, $\mathcal{D}(P) = \{\psi : \xi\hat{\psi} \in L^2\}$	`fourierL2`, `momentumOp`, `mem_domain_momentumOp`
$P\psi$ need not be defined	`roughState_not_mem_domain_momentumOp`, `domain_momentumOp_ne_top`
P and H_0 are self-adjoint	`isSelfAdjoint_momentumOp`, `isSelfAdjoint_freeHamiltonian`
$\mathcal{D}(P) = H^1(\mathbb{R})$, so $\mathcal{F}^{-1}\phi \notin H^1$	`mem_domain_momentumOp_iff_memSobolev`, `roughState_not_memSobolev`
$H_0 = P^2/2$ and e^{-itH_0}, defined on all of $L^2(\mathbb{R})$	`freeHamiltonian`, `freeEvolution` (from `L2.mulOp`, `L2.mulEvolution`)
on $\mathcal{D}(H_0)$ the Schrödinger equation holds	`hasDerivAt_freeEvolution` (from `L2.hasDerivAt_mulEvolution`)
off $\mathcal{D}(H_0)$ it holds at no time	`not_differentiableAt_freeEvolution`, `differentiableAt_freeEvolution_iff`
this evolution is that of Stone's theorem	`freeEvolution_eq_schrodingerPropagator`

Brief	Lean
Sec. 5, the harmonic oscillator in position space	
$H = P^2/2 + X^2/2$, diagonal in the Hermite basis	`hermiteUnitary`, `oscillatorHamiltonian` (from `TauCeti.hermiteHilbertBasis`)
H is self-adjoint, $H\psi_n = (n + \frac{1}{2})\psi_n$	`isSelfAdjoint_oscillatorHamiltonian`, `oscillatorHamiltonian_hermiteFunctionLp`
$H\psi_n = \frac{1}{2}(-\psi_n'' + x^2\psi_n)$	`oscillatorHamiltonian_hermiteFunctionLp_ae` (from `TauCeti.hermiteFunction_oscillator`)
the only self-adjoint operator with these eigenvectors	`eq_oscillatorHamiltonian` (from `lp.eq_diagonal`)
$\mathcal{D}(H) \neq L^2(\mathbb{R})$, and the Schrödinger equation holds only on $\mathcal{D}(H)$	`exists_lt_norm_oscillatorHamiltonian`, `oscillator_schrodinger_only_on_domain`

Brief	Lean
Sec. 5, the oscillator Hamiltonian as $H_0 + V$	
in $\mathcal{S}'$ and on Schwartz functions, $H_0 = -\frac{1}{2}\,d^2/dx^2$	`freeHamiltonian_toTemperedDistribution`, `freeHamiltonian_toLp`
$V = X^2/2$ is self-adjoint	`oscillatorPotential`, `isSelfAdjoint_oscillatorPotential`
$V\psi_n$ in terms of ψ_{n-2}, ψ_n, ψ_{n+2}	`sq_mul_hermiteFunction`, `oscillatorPotential_hermiteFunctionLp` (from `TauCeti.mul_hermiteFunction`)
$H_0\psi_n = (n + \frac{1}{2})\psi_n - V\psi_n$	`freeHamiltonian_hermiteFunctionLp`
$\mathcal{D}(H) \subseteq \mathcal{D}(H_0) \cap \mathcal{D}(V)$, and $H = H_0 + V$ there	`summable_smul_oscillatorPotential`, `oscillatorHamiltonian_le`
$H = H_0 + V$, with $\mathcal{D}(H) = \mathcal{D}(H_0) \cap \mathcal{D}(V)$	`oscillatorHamiltonian_eq_add`, `domain_oscillatorHamiltonian`
on Schwartz functions, $H\varphi = -\frac{1}{2}\varphi'' + \frac{1}{2}x^2\varphi$	`toLp_mem_domain_oscillatorHamiltonian`, `oscillatorHamiltonian_toLp`

Brief	Lean
Sec. 5, bounded potentials (V measurable, $\|V\| \le C$ a.e.)	
H_0 is unbounded on its domain	`exists_lt_norm_freeHamiltonian` (from `L2.exists_lt_norm_mulOp`)
$H = H_0 + V$ is self-adjoint, with $\mathcal{D}(H) = \mathcal{D}(H_0)$	`schrodingerHamiltonian`, `isSelfAdjoint_schrodingerHamiltonian`, `domain_schrodingerHamiltonian`
e^{-itH} solves the Schrödinger equation exactly on $\mathcal{D}(H) \neq L^2(\mathbb{R})$	`bounded_schrodinger_only_on_domain`

Brief	Lean
Sec. 5, potentials in $L^2 + L^\infty$ ($\|V\| \le \|W\| + C$ a.e., $W \in L^2$)	
on $L^2(\mathbb{R})$, $\mathcal{F}^{-1}f$ is the Fourier integral for integrable f	`Lp.fourierInv_coeFn_eq_of_integrable`, `Lp.ae_norm_fourierInv_le_of_integrable`
$\int \|f\| \le \varepsilon\|\xi^2 f\|_2 + C_\varepsilon\|f\|_2$	`Lp.integral_norm_le_of_ae_eq_sq_mul`, `Lp.exists_integral_norm_le_of_ae_eq_sq_mul`
$\|\psi\| \le \varepsilon\|H_0\psi\| + C_\varepsilon\|\psi\|$ a.e. on $\mathcal{D}(H_0)$	`exists_ae_norm_le_freeHamiltonian`
$\mathcal{D}(H_0) \subseteq \mathcal{D}(V)$, $\|V\psi\| \le a\|\psi\| + \frac{1}{2}\|H_0\psi\|$	`exists_norm_mulOp_le_freeHamiltonian`
Kato–Rellich: $H = H_0 + V$ is self-adjoint, $\mathcal{D}(H) = \mathcal{D}(H_0)$	`isSelfAdjoint_schrodingerHamiltonian_of_memLp`, `domain_schrodingerHamiltonian_of_memLp` (from `IsSelfAdjoint.add_of_relativeBound`)
e^{-itH} solves the Schrödinger equation exactly on $\mathcal{D}(H) \neq L^2(\mathbb{R})$	`exists_lt_norm_schrodingerHamiltonian_of_memLp`, `memLp_schrodinger_only_on_domain`

Brief	Lean
Sec. 5, real potentials ($V \in L^2_{\mathrm{loc}}(U)$, U open, U^c null)	
von Neumann: a densely defined symmetric operator commuting with a conjugation has a self-adjoint extension	`LinearPMap.exists_isSelfAdjoint_le_of_commutesWithConj`
$\mathcal{F}\bar{f}(\xi) = \overline{\mathcal{F}f(-\xi)}$ on $L^2(\mathbb{R})$	`Lp.fourier_conjugate`
$H_0 + V$ is symmetric and commutes with complex conjugation	`isFormalAdjoint_schrodingerHamiltonian`, `commutesWithConj_schrodingerHamiltonian`
$\mathcal{D}(H_0) \cap \mathcal{D}(V)$ is dense	`dense_domain_schrodingerHamiltonian`
$H_0 + V$ is unbounded on its domain	`exists_lt_norm_schrodingerHamiltonian_of_locallyIntegrableOn`
$H_0 + V$ has a self-adjoint extension H	`exists_isSelfAdjoint_le_schrodingerHamiltonian`
for every such H, e^{-itH} solves the Schrödinger equation exactly on $\mathcal{D}(H) \neq L^2(\mathbb{R})$	`realPotential_schrodinger_only_on_domain`
the same for continuous V, and for V continuous off 0	`continuous_schrodinger_only_on_domain`, `continuousOn_schrodinger_only_on_domain`

Brief	Lean
Sec. 5, commutator relations	
no bounded X, P with $[X, P] = i\hbar$	`not_exists_ccr_bounded` (from `commutator_ne_algebraMap`)
nor everywhere-defined symmetric ones	`not_exists_ccr_symmetric`
$[X, P]$ on its natural domain	`commutatorXP` (from `LinearPMap.compMax`)
$[X, P]$ is not defined on all of $L^2(\mathbb{R})$	`domain_commutatorXP_ne_top`
in $\mathcal{S}'$, X is multiplication by x and	`positionOp_toTemperedDistribution`,
$P = -i\, d/dx$	`momentumOp_toTemperedDistribution`
$[X, P]\psi = i\psi$ on $\mathcal{D}([X, P])$	`commutatorXP_apply` (from `TemperedDistribution.lineDerivOp_smulLeftCLM`)
$\mathcal{D}([X, P])$ contains the Schwartz functions, so is dense	`toLp_mem_domain_commutatorXP`, `dense_domain_commutatorXP`
all of the above	`commutator_relations`

Brief	Lean
Sec. 5, point-wise values	
$\psi \in L^2$ has representatives with any value at any point	`exists_representative_apply_eq` (from `MeasureTheory.update_ae_eq`)
so the values of ψ at a point are not uniquely defined	`exists_representatives_ne`
nor is the density $\lvert\psi(x)\rvert^2$	`exists_representative_density_eq`
point evaluation is not continuous in L^2, even on $\mathcal{S}(\mathbb{R})$	`not_exists_continuous_eval`
all of the above	`pointwise_values`

Differences from the brief.

- *Proposition 1.* Both parts are stated for linear maps, without Mathlib's partially defined operators. Discontinuity is proved for an unbounded linear map on any normed space, which includes $\mathcal{D}(A)$ with the norm inherited from $\mathcal{H}$. Hellinger–Toeplitz is proved for an everywhere-defined symmetric map; self-adjointness is not needed.

- *Proposition 2.* The number operator `numberOp` is the diagonal operator with entries n on $\ell^2(\mathbb{N})$, on its maximal domain, as a partially defined operator (`LinearPMap`); its domain is `numberDomain` $= \{x : \sum_n n^2 |x_n|^2 < \infty\}$. Self-adjointness is proved for every diagonal operator with real entries (`lp.diagonal`, in `ForMathlib`). The expectation `expN` $x = \sum_n n\,|x_n|^2/\|x\|^2$ is defined on all of ℓ^2 and equals $\mathrm{Re}\,\langle x|Nx\rangle\,/\|x\|^2$ on $\mathcal{D}(N)$ (`expN_eq_inner`). Discontinuity is proved within $\mathcal{D}(N)\setminus\{0\}$ with the subspace topology, as in the minimum form of [B]COBS. The sequence x_j is the brief's.

- *Proposition 3.* The wave functions are $\psi = (e^{-x^2}/\sqrt{\pi})^{1/2}$ (`gaussState`) and $\phi = (1/(\pi(1 + y^2)))^{1/2}$, as elements of $L^2(\mathbb{R})$. Both frames use the same space $L^2(\mathbb{R})$ with Lebesgue measure. The position operators are both multiplication by the coordinate, `positionOp`, on their common maximal domain `positionDomain` $= \{\psi : x\psi \in L^2\}$. Their self-adjointness is proved for every real measurable multiplication operator on $L^2(\mu)$ (`L2.isSelfAdjoint_mulOp`, in `ForMathlib`), by cutting off to the sets $\{|a| \le n\}$. That f is strictly increasing and onto $\mathbb{R}$,

and that $g = f^{-1}$ has $g' > 0$, are proved; smoothness is not, only differentiability, which is all the unitary needs. The unitary U is `Lp.compDiffeo`, proved unitary for any bijection g of $\mathbb{R}$ that is differentiable with a differentiable inverse, with weight $\sqrt{|g'|}$; its inverse V is the same construction for g^{-1}. The operator UXU^{-1} is the conjugate `conjLinearPMap`, with domain $U\,\mathcal{D}(X)$ by definition; that it is multiplication by $g(y)$ follows from $U(a\psi) = (a \circ g)\,U\psi$ for every measurable a (`Lp.compDiffeo_mulOp`). It differs from Y because $U\psi \in U\,\mathcal{D}(X)$ but $U\psi \notin \mathcal{D}(Y)$. Mathlib has no error function; `Real.erf` is defined in `ForMathlib`.

- *Proposition 4.* The classification of separable infinite-dimensional Hilbert spaces is new in `ForMathlib`. The positive integers n, m appear as $n+1, m+1$.

- *Theorem 5.* Stated as `theorem5`, a conjunction of three parts, one per condition, each for one particular system: the oscillator (Proposition 2), one spatial degree of freedom (Proposition 3), and any two numbers $n+1$, $m+1$ of continuous degrees of freedom (Proposition 4). "Not guaranteed" ("inconsistent" in the previous version of the brief) is read as the failure of the minimum form of the condition for that system. For [B]STDIST, which has no minimum form in the current brief, it is the one of the previous version, as the proof uses it. The [B]COBS part states that N is self-adjoint on $\mathcal{D}(N)$, that $E_N(x) = \operatorname{Re}\langle x|Nx\rangle / \|x\|^2$ there, and that E_N is not continuous on $\mathcal{D}(N) \setminus \{0\}$. The [B]WDQ part states $\psi \in \mathcal{D}(X)$, $U\psi = \phi$ a.e. and $U\psi \notin \mathcal{D}(Y)$, for the specific unitary U of the change of frame (for an arbitrary unitary the statement would be trivial). The [B]STDIST part states that a unitary $L^2(\mathbb{R}^{n+1}) \to L^2(\mathbb{R}^{m+1})$ exists and carries the QSTO:H model. For [B]STDIST, that a unitary carries rays, density operators, self-adjoint operators, the Born rule and the entropy is proved for any unitary between complex Hilbert spaces (`carriesQSTO`). Rays are elements of `Projectivization`; density operators and the entropy are those of `ForPhyslib` (positive, trace class, trace one; entropy in $[0, \infty]$); self-adjoint operators are partially defined (`LinearPMap`).

- *Theorem 5, Schwartz spaces.* "Schwartz spaces present the same problem" is read in its strongest form: for all $n, m \geq 1$ there is a unitary $U\colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^m)$ that carries the QSTO:H model and maps $\mathcal{S}(\mathbb{R}^n)$ onto $\mathcal{S}(\mathbb{R}^m)$, as an isomorphism of topological vector spaces (`schwartz_stdist`). So the rigged Hilbert spaces $\mathcal{S} \subseteq L^2$ do not tell the systems apart either. As for [B]STDIST, U does not carry the positions and momenta. The proof is the Hermite characterization of $\mathcal{S}$ (the N-representation of Reed–Simon I, Sec. V.3 and its appendix): $f \in L^2(\mathbb{R}^n)$ is a Schwartz function exactly when $\sup_\alpha (1+|\alpha|)^K |c_\alpha(f)| < \infty$ for all K, and these suprema define the topology of $\mathcal{S}$. One direction uses the operator $\sum_i (x_i + \partial_i)(x_i - \partial_i) + 2$, which multiplies c_α by $2(|\alpha| + 1 + n)$, the other the Sobolev bound for the values, Parseval's identity and the ladder relations. The unitary U relabels the Hermite functions, $h_\alpha \mapsto h_{\sigma(\alpha)}$, by a bijection $\sigma\colon \mathbb{N}^n \to \mathbb{N}^m$ built from iterated Cantor pairings, for which $1+|\sigma(\alpha)|$ and $1+|\alpha|$ are polynomially bounded by each other.

- *Theorem 5, classical cutoffs.* The brief contrasts [B]COBS and [B]WDQ with classical mechanics, where "one can, in principle, put a cutoff on all variables" and work with "joint probability distributions with compact support, which bound all moments", while "Quantum mechanics does not allow that". Classically the states are read as the probability measures on phase space $\mathbb{R}^n \times \mathbb{R}^n$ concentrated on a compact set K, with the topology of weak convergence. Every continuous observable then has a finite expectation, continuous in the state (`classical_cutoff_cobs`): it agrees on K with a bounded continuous function, obtained by clamping. A continuous change of frame φ maps such a state to one concentrated on the compact set $\varphi(K)$, so all moments stay finite (`classical_cutoff_wdq`). In quantum mechanics a cutoff on all variables cuts off position and momentum, and the only such state is

zero (`quantum_no_cutoff`, for one degree of freedom): if $\psi \in L^2(\mathbb{R})$ vanishes outside $(a, b]$, then $G(z) = \int_a^b e^{-2\pi i x z}\psi(x)\,dx$ is entire and agrees with $\mathcal{F}\psi$ on $\mathbb{R}$; if $\mathcal{F}\psi$ also vanishes outside a bounded set, the identity theorem gives $G = 0$, so all Fourier coefficients of ψ on $[a, b]$ vanish, and $\psi = 0$ by Parseval's identity. A cutoff on position alone is allowed, and is not formalized: smooth compactly supported states have finite moments of all orders in position and momentum.

- *Sec. 5, the domain of the Schrödinger equation.* The brief says that the equation is valid only on $\mathcal{D}(H)$. This is read dynamically: the evolution e^{-itH} is defined on all of $\mathcal{H}$, but its orbit $t \mapsto e^{-itH}\psi$ satisfies $i\frac{d}{dt}\psi(t) = H\psi(t)$ exactly when $\psi \in \mathcal{D}(H)$, and is differentiable at no time otherwise. It is proved for every self-adjoint H that is unbounded on its domain (`schrodinger_only_on_domain`): $\mathcal{D}(H) \neq \mathcal{H}$ by Hellinger–Toeplitz, and $e^{-itH/\hbar}$ is the evolution of Stone's theorem (`schrodingerPropagator`, in `ForPhyslib`, from PhyslibAlpha's spectral theorem for unbounded self-adjoint operators), whose orbits are differentiable exactly when they start in $\mathcal{D}(H)$ (`differentiableAt_schrodingerPropagator_iff`, new in `ForPhyslib`). This covers every potential V for which $P^2/2m + V$ is self-adjoint; that it is, is proved here for $V = 0$, for bounded measurable V, for $V \in L^2 + L^\infty$, and for the oscillator, $V(x) = x^2/2$; for every other real potential in L^2_{loc} off a closed null set, it has self-adjoint realizations. The evolution is also built explicitly, and shown to be the one of Stone's theorem, for the harmonic oscillator in the number basis, $H = N$ on $\ell^2(\mathbb{N})$, in units with $\hbar\omega = 1$ and without the zero-point energy, which only adds a global phase. The explicit evolution and the statement are proved for every real diagonal operator on ℓ^2 (`lp.diagonalEvolution`, in `ForMathlib`), and for every real multiplication operator on $L^2(\mu)$ (`L2.mulEvolution`). In position space, the momentum operator is $P = \mathcal{F}^{-1}(2\pi\xi)\mathcal{F}$ with Mathlib's unitary Fourier transform on $L^2(\mathbb{R})$, whose convention $\hat{\psi}(\xi) = \int e^{-2\pi i x\xi}\psi(x)\,dx$ gives the factor 2π ($\hbar = 1$). Its maximal domain is identified with $H^1(\mathbb{R})$ in Mathlib's definition (`MemSobolev 1 2`, through the Bessel potential); that this is the space of L^2 functions whose distributional derivative is in L^2 is not proved here. The state outside $\mathcal{D}(P)$ is $\mathcal{F}^{-1}\phi$, for the Cauchy packet ϕ of Proposition 3. The Schrödinger equation is checked in position space for the free particle, $H_0 = \mathcal{F}^{-1}(2\pi^2\xi^2)\mathcal{F}$ (mass 1); that $H_0 = P^2/2$ as partially defined operators is not proved. The harmonic oscillator in position space ($m = \omega = \hbar = 1$) is $H = \sum_n (n + \frac{1}{2})\,|\psi_n\rangle\langle\psi_n|$ on its maximal domain, with TauCeti's Hermite functions ψ_n and their Hilbert basis of $L^2(\mathbb{R})$. It is tied to the differential expression $\frac{1}{2}(-\psi'' + x^2\psi)$ by $H\psi_n = \frac{1}{2}(-\psi_n'' + x^2\psi_n)$, and by uniqueness: H is the only self-adjoint operator with these eigenvectors and eigenvalues (`eq_oscillatorHamiltonian`, from `lp.eq_diagonal`, new in `ForMathlib`). It is also the operator sum $H = H_0 + V$ of partially defined operators, with $V = X^2/2$ multiplication by $x^2/2$, on $\mathcal{D}(H) = \mathcal{D}(H_0) \cap \mathcal{D}(V)$ (`oscillatorHamiltonian_eq_add`). The proof computes H_0 on Schwartz functions in $\mathcal{S}'$, and $V\psi_n$ from the position ladder relation; for $x \in \mathcal{D}(H)$ the series $\sum_n c_n V\psi_n$ then converges, so $x \in \mathcal{D}(V)$ and $x \in \mathcal{D}(H_0)$ because both operators are closed. Hence $H \subseteq H_0 + V$, and equality holds because $H_0 + V$ is symmetric and H is self-adjoint (`LinearPMap.eq_of_le_of_isFormalAdjoint`, new in `ForMathlib`). In particular $H\varphi = -\frac{1}{2}\varphi'' + \frac{1}{2}x^2\varphi$ for every Schwartz function φ (`oscillatorHamiltonian_toLp`). For a measurable V with $|V| \leq C$ almost everywhere, $H_0 + V$ is self-adjoint on $\mathcal{D}(H_0)$ (`isSelfAdjoint_schrodingerHamiltonian`). This is the bounded case of the Kato–Rellich theorem, proved directly: multiplication by V is defined on all of L^2 and symmetric, and a self-adjoint operator plus an everywhere defined symmetric one is self-adjoint (`IsSelfAdjoint.add_of_domain_eq_top`, new in `ForMathlib`). Since H_0 is unbounded on its domain (`exists_lt_norm_freeHamiltonian`), so is $H_0 + V$, and the remark holds for it (`bounded_schrodinger_only_on_domain`).

More generally, let $|V| \le |W| + C$ almost everywhere with $W \in L^2$, that is $V \in L^2 + L^\infty$; this includes singular potentials such as $|x|^{-\alpha}$ near 0 for $\alpha < \frac{1}{2}$. Then $H_0 + V$ is self-adjoint on $\mathcal{D}(H_0)$ (`isSelfAdjoint_schrodingerHamiltonian_of_memLp`), by the Kato–Rellich theorem. The relative bound comes from the one-dimensional Sobolev bound: for $\psi \in \mathcal{D}(H_0)$, $f = \mathcal{F}\psi$ and $\xi^2 f$ are in L^2, so by Cauchy–Schwarz with $(\lambda+\xi^2)^{-1} \in L^2$, $\int |f| \le \sqrt{\pi/\lambda}\,(\lambda\|f\|_2 + \|\xi^2 f\|_2)$ for $\lambda \ge 1$ (`Lp.integral_norm_le_of_ae_eq_sq_mul`). Mathlib's L^2 Fourier transform is defined by extension from Schwartz functions; that on an integrable f the inverse transform is the Fourier integral is new (`Lp.fourierInv_coeFn_eq_of_integrable`, through tempered distributions and the uniqueness of distributions given by locally integrable functions). Hence $|\psi| \le \int |f| \le \varepsilon\|H_0\psi\| + C_\varepsilon\|\psi\|$ almost everywhere, and $\|V\psi\| \le \|W\|_2\,\|\psi\|_\infty + C\|\psi\| \le a\|\psi\| + \frac{1}{2}\|H_0\psi\|$. The remark of Sec. 5 follows as for bounded V (`memLp_schrodinger_only_on_domain`).

TauCeti has the Kato–Rellich theorem (`IsSelfAdjoint.add_of_isRelativelyBounded`), but its module cannot be imported together with Physlib's unbounded operators, which Stone's theorem here uses: both declare `LinearPMap.adjoint_smul`. The theorem is reached through `ForMathlib/Analysis/InnerProductSpace/KatoRellich.lean`, which restates it with Mathlib's notions only (`IsSelfAdjoint.add_of_relativeBound`) and imports TauCeti privately; modules importing it do not load TauCeti. Files that load private imports, such as non-module files and the exporter of `lake comparator`, still see the clash, so these results are in the separate module `HilbertUnphysicality.KatoRellich`, which the comparator lane does not import.

For potentials outside $L^2 + L^\infty$, such as $|x|^\beta$ with $\beta > 0$, $-x^4$ or $\pm 1/x^2$, $P^2/2m + V$ is not H_0-bounded, and need not be self-adjoint on $\mathcal{D}(H_0) \cap \mathcal{D}(V)$ nor have a unique self-adjoint extension. The brief speaks of "the" Hamiltonian; it is read as any self-adjoint realization: a self-adjoint H extending $H_0 + V$ on $\mathcal{D}(H_0) \cap \mathcal{D}(V)$. Every real V that is square integrable near each point of an open set U with null complement has one (`exists_isSelfAdjoint_le_schrodingerHamiltonian`), and the remark holds for every one (`realPotential_schrodinger_only_on_domain`); this includes every continuous V and every measurable V continuous off 0. The domain $\mathcal{D}(H_0) \cap \mathcal{D}(V)$ is dense, since a vector orthogonal to it is orthogonal to every smooth function with compact support in U, and so vanishes on U. The operator commutes with complex conjugation: multiplication by a real V does, and so does H_0, the Fourier transform of multiplication by the even function $2\pi^2\xi^2$, because $\mathcal{F}\bar{f}(\xi) = \overline{\mathcal{F}f(-\xi)}$ (`Lp.fourier_conjugate`, new, by density of Schwartz functions). By von Neumann's theorem it has a self-adjoint extension; the theorem is new in `ForMathlib` (`LinearPMap.exists_isSelfAdjoint_le_of_commutesWithConj`). The proof takes, by Zorn's lemma, a maximal symmetric extension S commuting with the conjugation C. It is closed, as S^{**} is another one. If $S^*k = ik$ with $k \ne 0$, then $S^*Ck = -iCk$, and S extends to $\mathcal{D}(S) + \operatorname{span}(k + Ck)$, which is excluded; so the deficiency spaces are zero, and S is self-adjoint by TauCeti's criterion, reached through a private import as for Kato–Rellich (`isSelfAdjoint_of_surjective_addSub`). The extension is unbounded, since already $H_0 + V$ is (`exists_lt_norm_schrodingerHamiltonian_of_locallyIntegrableOn`): for a bump f with support in U, multiplying by e^{itx} changes neither $\|f\|$ nor $\|Vf\|$, while

$$\|H_0\, e^{itx} f\| \ge \tfrac{1}{2}\,(t^2\|f\| - 2t\|f'\| - \|f''\|).$$

The self-adjoint realization is not unique in general: for $-x^4$ or $-1/x^2$ they differ by boundary conditions at infinity or at 0. Which one is physical is a choice of the model, not a condition of the remark, which holds for each. For the potential part, $V(y) = y^2$ and the state $\phi = U\psi$

of Proposition 3 are used: $\phi \in L^2$, but $V\phi \notin L^2$; for $V(y) = y$ this is `frameUnitary_gaussState_not_mem`.

- *Sec. 5, commutator relations.* The brief says that $[X, P]/(i\hbar) = I$ cannot hold on the whole of $\mathcal{H}$, since the left side is unbounded, and must be restricted to a domain. Both halves are proved, with $\hbar = 1$. The commutator `commutatorXP` is $XP - PX$ on its natural domain $\{\psi \in \mathcal{D}(X) \cap \mathcal{D}(P) : P\psi \in \mathcal{D}(X),\ X\psi \in \mathcal{D}(P)\}$. Mathlib composes partially defined operators only when the range of one lies in the domain of the other, so the composition on the natural domain is new in `ForMathlib` (`LinearPMap.compMax`). That the relation cannot hold everywhere is proved in two forms: $\mathcal{D}([X, P]) \neq L^2(\mathbb{R})$; and, on any nonzero Hilbert space, no everywhere-defined symmetric X, P satisfy $XP - PX = i\hbar I$. The second form is the Wielandt–Wintner theorem (in a nontrivial normed algebra no commutator is a nonzero scalar, `commutator_ne_algebraMap`, in `ForMathlib`) combined with Hellinger–Toeplitz. That the restricted relation holds is proved on all of $\mathcal{D}([X, P])$: in the tempered distributions, X is multiplication by x and $P = -i\, d/dx$, the Leibniz rule $\frac{d}{dx}(xu) = u + x\frac{du}{dx}$ holds (`TemperedDistribution.lineDerivOp_smulLeftCLM`, in `ForMathlib`), and the embedding $L^2 \to \mathcal{S}'$ is injective. Schwartz functions lie in $\mathcal{D}([X, P])$, so it is dense. Closures and self-adjoint extensions are not considered.

- *Sec. 5, point-wise values.* The brief says that a vector of L^2 is a class of functions that agree up to a null set, so its values at a point, and the density $|\psi(x)|^2$, are not uniquely defined. This is proved for $L^2(\mathbb{R})$: since points are null, changing a representative at one point gives another representative (`MeasureTheory.update_ae_eq`, in `ForMathlib`), so every value at every point is taken, by ψ and by $|\psi|^2$. Beyond the brief, the value at a point cannot be fixed by a continuous choice either: no continuous linear functional on $L^2(\mathbb{R})$ agrees with $\varphi \mapsto \varphi(x_0)$ on the Schwartz functions (`not_exists_continuous_eval`). The witnesses are smooth bumps of value 1 at x_0 whose L^2 norm tends to 0. That $|\psi(x)|^2$ is "conceptually uniquely defined" is the physical intuition the brief contrasts with, and is not a mathematical claim.

Problems found in the brief.

- Cross-references lack their label: the proof of Theorem 5 has "This satisfies the conditions for 2", "As shown in 3" and "By 4" for Propositions 2, 3 and 4.

- "Schwartz spaces present the same problem" is asserted without proof, and without saying which structure on $\mathcal{S}(\mathbb{R}^n)$ is meant. The formalization proves it in its strongest form (see "Differences").

- QSTO:H still allows any separable Hilbert space, finite-dimensional ones included. This does not affect Theorem 5, which only claims that the conditions are not guaranteed, that is, that each fails for some system.

- The current brief states [B]STDIST as "Systems with a different number of degrees of freedom are distinct", which does not say in what sense they are distinct. The minimum form of the previous version, that they "should not have the same mathematical representation", was dropped, but the proof of Theorem 5 still concludes that they have "the same mathematical representation". The formalization keeps that minimum form.

- The remarks on classical cutoffs in the proof of Theorem 5 are informal: neither the classical states with a cutoff nor the sense in which quantum mechanics does not allow one is specified. The formalization proves both in the readings of the open problems (see "Differences").